

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Sixth Semester Theory B. Tech (ME) Examination****May 2018****ME 310 Production Technology****Date: 08.05.2018, Tuesday****Time: 10:00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions.**

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| 1. | Define the terms Machinability and Machinability index. | 01 |
| 2. | Write Tool signature according to ASA notation given as : 16-12-8-8-10-15-3 | 01 |
| 3. | What do you mean by chips with built up edge formation? | 01 |
| 4. | Under certain cutting conditions, doubling the cutting speed reduces the tool life to $(1/16)^{\text{th}}$ of the original. Taylor's tool life index (n) for this tool workplace combination will be ____. | 02 |

Q – 2 Answer any four questions.**16**

1. State the difference between orthogonal cutting and oblique cutting with neat sketch.
2. What are flank wear and crater wear? Explain briefly with sketch.
3. Name several techniques for determining tool chip contact temperature. Describe the tool/work thermocouple method for temperature measurement in brief.
4. Identify the category of parallel processing type of lathe machine. How it differs from progressive processing type of turret lathe machine?
5. Enlist various requirements of semi automats and automats in today's life.
6. Jay is an entrepreneur. He owns a company which makes tailor made machine tools as per requirements of his customers. He has some trouble while selecting the material for its machine tool structure. Suggest him some points to keep in mind during selection of material for his machine tool structure.

Q – 3 Answer any two questions.**14**

1. During an orthogonal cutting of mild steel with a carbide tool at Miranda machine tools pvt. Ltd. the various parameters were noted as ,
 - rake angle was 10°
 - the cut width was 3 mm
 - the feed was set at 0.4 mm/rev.
 - the chip thickness was measured to be 0.7 mm
 - the cutting speed was 120 m/min.
 - the forces measured by dynamometer were $F_z = 1800$ N and $F_{xy} = 1200$ N
 Determine the shear force F_s , compressing force F_c and coefficient of friction μ using a graphical approach of merchant circle diagram (MCD) .
 Also determine the shear velocity and chip flow velocity from velocity diagram.
2. Draw a neat sketch of single point cutting tool with label of six major angles and other terminology of it. Differentiate the significance of providing positive and negative rake surfaces to the single point cutting tool.
3. Find the speed steps arranged in geometric and harmonic progression for the following conditions: $n_{\min} = 12$ rpm , $n_{\max} = 510$ rpm, $z = 8$, $v = 10$ m/ min.

SECTION – II**Q - 4 Answer the following questions.****07**

1. In electro discharge machining, tool is made of

a) brass	b) copper	c) copper tungsten alloy	d) all of these
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2. Chemical milling operation is performed

a) on universal milling machine	b) on plain milling machine	c) in a tank containing an etching solution	d) any one of these
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3. In hot machining, tool is made of

a) tungsten carbide	b) brass or copper	c) diamond	d) stainless steel
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4. In ultrasonic machining, the metal is removed by
 - a) using abrasive slurry between the tool and work
 - b) direct contact of tool with the work
 - c) maintaining an electrolyte between the work and tool in a very small gap between the two
 - d) erosion caused by rapidly recurring spark discharges between the tool and work

5. Gear lapping is an operation

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|-------------------------|----------------------------|----------------------------|------------------|
| a) after heat treatment | b) prior to heat treatment | c) for gear reconditioning | d) none of these |
|-------------------------|----------------------------|----------------------------|------------------|

6. The use of jigs and fixtures

- a) facilitates deployment of less skilled labour for production
- b) eliminates pre-machining operations like marking, measuring, laying out etc.
- c) reduced manual handling operations
- d) all of the above

7. A fixture is defined as a device which

- a) holds and locates a workpiece and guides and controls one or more cutting tools
- b) holds and locates a workpiece during an inspection or for a manufacturing operation
- c) is used to check the accuracy of workpiece
- d) all of the above

Q - 5 (a) What do you understand by the term “Modern Machining Processes”? What is their importance? Describe parameters considered for selection of modern machining process. **07**

(b) Describe basic principle of ECG with neat sketch. What are the differences between ECG and conventional grinding? What is the role of electrolyte in ECG process? **07**

OR

Q - 5 (a) Describe principle of Abrasive Jet Machining (AJM) process. Differentiate between Abrasive Jet Machining Process (AJM) and Water Jet Machining (WJM) Process. **07**

(b) Explain basic principle of EDM with neat sketch. Justify the use of servo control system in EDM. **07**

Q - 6 (a) Describe different methods used for location with sketch. **07**

(b) Illustrate the process of thread cutting on centre lathe. Discuss the thread grinding operation in brief. **07**

OR

Q - 6 (a) Explain construction and purpose of Template jigs, Plate jigs, Table jig and Channel jigs. **07**

(b) Describe the process of gear cutting by gear shaper with neat sketch. Discuss the importance of cutting speed, indexing motion and depth of cut for the process. **07**
